Deliverable 1.8 — Boost Converter IC Selection

EE499 Final Year Project DC-DC Boost Converter for Voltage Stabilisation in Battery-Powered Robotic Subsystems Fortune Olose · Maynooth University · Supervisor: Dr. Bob Lawlor

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1. Overview

This deliverable evaluates and selects the boost converter integrated circuit (IC) that will serve as the controller for the power stage. The IC is the central component of the converter — it generates the gate drive signals, implements the control loop, and provides the protection features (overcurrent, overvoltage, thermal shutdown) that make the converter safe and reliable.

Three candidate ICs are evaluated against the design requirements established in Deliverables 1.1–1.7. The candidates span a range of integration levels: a fully integrated synchronous converter with internal power switches, an external-MOSFET synchronous controller, and a low-cost non-synchronous converter with an internal switch. This range allows a meaningful engineering comparison between design simplicity, efficiency, flexibility, and cost.

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2. Design Requirements for IC Selection

The following requirements are derived from the specifications established in prior deliverables:

Requirement	Value	Source
Input voltage range	8–11 V	Deliverable 1.1
Output voltage	12 V (regulated)	Deliverable 1.1
Maximum output current	3 A	Deliverable 1.1
Maximum output power	36 W	Deliverable 1.1
Switching frequency	≥ 200 kHz (adjustable preferred)	Deliverable 1.2

Peak inductor current (worst case)	~4.95 A	Deliverable 1.4/1.6
Control mode	Current-mode preferred	Design preference
Rectification	Synchronous preferred (higher η)	Design preference
Package	SMD, hand-solderable preferred	Prototyping constraint

Additionally, the following qualitative factors are considered:

- **Documentation quality** — availability of detailed datasheets, application notes, reference designs, and evaluation modules (EVMs).
- **Simulation model availability** — SPICE or TINA-TI models for LTspice validation in Phase 2.
- **Supply chain availability** — stock at Mouser, Farnell, or Digi-Key for reliable procurement.
- **Design complexity** — number of external components required, ease of PCB layout.

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3. Candidate ICs

3.1 Candidate A: Texas Instruments TPS61088

Type: Fully integrated synchronous boost converter **Datasheet:** TI document SLVSC30D (Rev. D, August 2021)

The TPS61088 is a high-power-density synchronous boost converter with both the low-side and high-side power MOSFETs integrated on-chip. The internal switches have R_DS(on) values of 11 mΩ (low-side) and 13 mΩ (high-side), which eliminates the need for external power MOSFETs and a Schottky diode entirely.

Key Specifications:

Parameter	Value
Input voltage range	2.7–12 V
Output voltage range	Up to 12.6 V
Maximum switch current	10 A
Switching frequency	200 kHz – 2.2 MHz (resistor-programmable)

Control topology	Adaptive constant off-time, peak current mode
Internal low-side R_DS(on)	11 mΩ
Internal high-side R_DS(on)	13 mΩ
Quiescent current (PFM mode)	~30 µA
Output power capability	Up to 30 W (from datasheet; dependent on V_in)
Protections	OVP (13.2 V), cycle-by-cycle OCP, thermal shutdown (150°C)
Package	4.50 × 3.50 mm, 20-pin VQFN
Light-load mode	PFM or forced-PWM (MODE pin selectable)
Recommended inductor range	0.47–10 µH

Strengths:

- Fully integrated — no external MOSFETs or diode required, minimising BOM count and PCB area.
- Synchronous rectification provides higher efficiency than a Schottky diode (eliminates ~0.4 V forward drop loss).
- Adjustable switching frequency from 200 kHz to 2.2 MHz via a single external resistor, providing design flexibility.
- Peak current mode control offers inherent line feed-forward, simplified loop compensation, and cycle-by-cycle current limiting.
- Excellent documentation: detailed 35-page datasheet, multiple TI reference designs (PMP9772, PMP9778, PMP9779), EVM available (TPS61088EVM-638), and TINA-TI/PSpice simulation models.
- 10 A switch current capability provides substantial headroom above the ~4.95 A peak inductor current in this design.
- Widely available from authorised distributors (Mouser, Digi-Key, Farnell).

Limitations:

- Maximum output voltage of 12.6 V. For the 12 V target, this provides only 0.6 V of headroom. While sufficient for regulation, it limits any future extension to higher output voltages.
- Recommended inductor range is 0.47–10 µH. The 15 µH inductor calculated in Deliverable 1.4 at 200 kHz falls outside this range. This necessitates either increasing the switching frequency (to bring the required inductance within range) or accepting a 10 µH inductor with slightly higher ripple current. This interdependency is addressed in Section 5.

- The 20-pin VQFN package with exposed thermal pad requires reflow soldering or careful hot-air rework — not trivially hand-solderable, though manageable with a heat gun and stencil.
- Output power capability is V_in-dependent. At V_in = 8 V with a 12 V output, the available output power may be limited to approximately 20–25 W due to the peak switch current limit, which is close to the 24 W nominal specification. Thermal management and current limit programming must be carefully considered.

3.2 Candidate B: Texas Instruments LM5122

Type: Synchronous boost controller (external MOSFETs required) **Datasheet:** TI document SNVS954H (Rev. H, June 2017)

The LM5122 is a wide-input synchronous boost controller that drives external N-channel MOSFETs for both the low-side and high-side switches. It provides the gate drive, control loop, and protection logic, but the power stage components are selected independently by the designer.

Key Specifications:

Parameter	Value
Input voltage range	3–65 V (4.5 V for start-up)
Output voltage range	Up to 100 V
Switching frequency	Up to 1 MHz (resistor-programmable or synchronisable)
Control topology	Peak current mode
Gate driver current	3 A peak (both high-side and low-side)
Adaptive dead-time control	Yes
Multi-phase capability	Yes (180° phase-shifted clock output)
Protections	Cycle-by-cycle OCP, hiccup-mode overload, thermal shutdown, UVLO
Package	4.00 × 4.00 mm, 20-pin WQFN
Diode emulation mode	Optional (MODE pin)
Bypass (V_out = V_in) operation	Yes, via internal BST charge pump

Strengths:

- Extremely wide input (3–65 V) and output (up to 100 V) voltage range — far exceeds the present design requirements, offering maximum flexibility for future applications or modified specifications.

- External MOSFET selection allows the designer to independently optimise the power stage for efficiency, thermal performance, and voltage/current ratings. This is a significant educational advantage — the designer must understand MOSFET gate charge, R_DS(on), and switching loss tradeoffs.
- Peak current mode control with robust 3 A gate drivers and adaptive dead-time control.
- Multi-phase capability for higher-power applications (not needed here, but demonstrates scalability).
- No restriction on inductor value — the 15 µH inductor from Deliverable 1.4 is fully compatible.
- Comprehensive TI documentation, EVM (LM5122EVM-1PH), WEBENCH design tool support, and TINA-TI models.

Limitations:

- Requires external MOSFETs (low-side and high-side N-channel), gate drive bootstrap capacitor, and current sense resistor. This significantly increases the BOM count (approximately 8–12 additional power-path components compared to the TPS61088).
- PCB layout complexity increases with external power stage — gate drive loop inductance, switch-node ringing, and thermal management all require careful attention.
- The 4.5 V minimum start-up voltage is above the 2.7 V minimum of the TPS61088, though this is irrelevant for the 8–11 V input specification.
- Higher total solution cost due to external MOSFETs, sense resistor, and bootstrap components.
- Significant over-engineering for a 12 V / 3 A application — this controller is designed for much higher power levels (the EVM demonstrates 24 V / 4.5 A output from 9–20 V input).

3.3 Candidate C: XLSEMI XL6009E1

Type: Non-synchronous boost converter with internal N-channel MOSFET **Datasheet:** XLSEMI XL6009E1 Rev 1.1

The XL6009E1 is a low-cost, current-mode boost converter with a built-in 4 A N-channel MOSFET switch. It is the successor to the legacy LM2577 and is widely used in low-cost boost converter modules available from online retailers.

Key Specifications:

Parameter	Value
Input voltage range	5–32 V (datasheet; modules claim 3–32 V)
Output voltage range	Up to 60 V (with external diode)
Maximum switch current	4 A

Switching frequency	400 kHz (fixed)
Control topology	Current mode
Internal MOSFET	N-channel, integrated
Rectification	Non-synchronous (external Schottky diode required)
Protections	OVP on SW pin, thermal shutdown, current limiting
Package	TO-263-5 (D2PAK)
Reference voltage	1.25 V
Efficiency	Up to 94% (claimed; heavily dependent on operating conditions)

Strengths:

- Very low cost (< €1 per unit in quantity; pre-built modules available for ~€1–2).
- Simple application circuit — few external components required beyond the diode, inductor, and feedback resistors.
- TO-263-5 package is large and easy to hand-solder, ideal for prototyping.
- Fixed 400 kHz frequency eliminates the need for a frequency-setting resistor.
- 1.25 V reference with external resistor divider provides straightforward output voltage programming.

Limitations:

- **4 A maximum switch current is critically marginal.** The worst-case peak inductor current in this design is ~4.95 A (at V_in = 8 V, I_out = 3 A). This exceeds the XL6009's absolute maximum switch current rating, meaning the IC will current-limit or enter protection before the converter reaches its maximum load specification. This is a disqualifying limitation.
- Non-synchronous rectification requires an external Schottky diode, which introduces ~0.3–0.5 V forward drop loss. At 3 A output, the diode dissipates 0.9–1.5 W, reducing efficiency by approximately 3–5% compared to a synchronous design.
- Fixed 400 kHz switching frequency provides no flexibility for optimising the efficiency/inductor-size tradeoff.
- Relatively poor documentation — the datasheet is only 8 pages with limited application guidance, no official SPICE model, no reference designs, and no EVM.
- The manufacturer (XLSEMI) is a smaller Chinese semiconductor company with less comprehensive long-term support and validation compared to Texas Instruments.
- Real-world performance often falls significantly short of datasheet claims. Independent testing suggests that at wide boost ratios (e.g., 3 V → 12 V), the XL6009 may struggle to deliver even a few hundred milliamps continuously without thermal runaway.

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4. Comparative Summary

Parameter	TPS61088	LM5122	XL6009E1
Type	Integrated sync converter	External-FET sync controller	Integrated non-sync converter
V_in range	2.7–12 V	3–65 V	5–32 V
V_out max	12.6 V	100 V	60 V
I_switch max	10 A	External (unlimited)	4 A
f_sw	200 kHz – 2.2 MHz	Up to 1 MHz	400 kHz (fixed)
Control mode	Peak current (const off-time)	Peak current	Current mode
Synchronous	Yes (internal)	Yes (external FETs)	No
Inductor range	0.47–10 μH	No restriction	No restriction
External FETs needed	No	Yes (2× N-channel)	No
External diode needed	No	No	Yes (Schottky)
BOM complexity	Low (~15 components)	High (~25+ components)	Low (~10 components)
PCB area	Small	Large	Medium
Documentation	Excellent (TI)	Excellent (TI)	Poor (8-page DS)
SPICE model	Yes (PSpice/TINA-TI)	Yes (TINA-TI)	No
EVM available	Yes	Yes	No
Peak I_L headroom	10 A >> 4.95 A ✓	External — designer's choice ✓	4 A < 4.95 A ✗
Approx. unit cost	~€3–5	~€3–4 (IC only; + FET cost)	< €1
Suitability	★★★★★	★★★☆☆	★★☆☆☆

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5. Selection: TPS61088

5.1 Rationale

The **Texas Instruments TPS61088** is selected as the boost converter IC for this design. The selection is based on the following primary factors:

1. **Full integration eliminates the power stage design problem.** With both MOSFETs integrated on-chip (11 mΩ and 13 mΩ R_DS(on)), the TPS61088 removes the need to select, characterise, and lay out external power transistors and a Schottky diode. For a final year project with a constrained timeline, this significantly reduces PCB design risk and prototyping iteration cycles.

2. **Synchronous rectification maximises efficiency.** The integrated high-side MOSFET replaces a Schottky diode, eliminating the ~0.4 V forward drop that would otherwise dissipate 1.2 W at 3 A output. This is a direct, measurable efficiency improvement that will be quantified in simulation (Phase 2) and hardware testing (Phase 4).

3. **10 A switch current provides generous headroom.** The worst-case peak inductor current of ~4.95 A uses less than 50% of the TPS61088's switch current capability, providing robust margin for transient events and component tolerances.

4. **Adjustable switching frequency enables design optimisation.** The 200 kHz to 2.2 MHz range allows the switching frequency to be tuned during the design process — a critical advantage given the inductor range consideration discussed in Section 5.2.

5. **Excellent ecosystem for simulation and validation.** TI provides PSpice/TINA-TI transient models, the TPS61088EVM-638 evaluation module, and multiple reference designs (PMP9772, PMP9773, PMP9778, PMP9779) that demonstrate proven layouts and component selections. This accelerates the simulation phase and provides validated reference points for PCB design.

6. **Peak current mode control simplifies compensation.** The adaptive constant off-time topology with peak current sensing provides inherent line feed-forward (output voltage is naturally resistant to input voltage changes) and simplifies the feedback loop design, which is beneficial for a student project where loop compensation expertise may be developing.

5.2 Design Iteration: Inductor Value and Switching Frequency

The TPS61088 datasheet specifies a recommended inductor range of **0.47–10 µH**. The 15 µH inductor calculated in Deliverable 1.4 was derived assuming f_sw = 200 kHz. Since the TPS61088's adaptive constant off-time control architecture is optimised for the recommended inductor range, the switching frequency should be increased to bring the required inductance within this range.

Revised calculation at f_sw = 350 kHz:

At V_{in} = 8 V, D = 0.333, I_{out} = 2 A:

$I_{L_avg} = I_{out} / (1 - D) = 2 / 0.667 = 3.0$ A
$\Delta I_L = 30\% \times I_{L_avg} = 0.9$ A

$L = (V_{in} \times D) / (\Delta I_L \times f_{sw})$
$= (8 \times 0.333) / (0.9 \times 350 \times 10^3)$
$= 2.664 / 315{,}000$
$= 8.46$ μH → round to 10 μH (standard value)

A **10 μH inductor at 350 kHz** falls within the TPS61088's recommended range and remains a standard, readily available value. The ripple current at this operating point becomes:

$\Delta I_L = (V_{in} \times D) / (L \times f_{sw})$
$= (8 \times 0.333) / (10 \times 10^{-6} \times 350 \times 10^3)$
$= 2.664 / 3.5$
$= 0.761$ A

This is 25.4% of I_{L_avg} — slightly lower than the original 30% target, which is acceptable and actually beneficial (lower ripple reduces output voltage ripple and core losses).

Updated CCM boundary at 350 kHz, L = 10 μH:

$I_{out_min_CCM} = (V_{in} \times D \times (1 - D)) / (2 \times L \times f_{sw})$
$= (8 \times 0.333 \times 0.667) / (2 \times 10 \times 10^{-6} \times 350 \times 10^3)$
$= 1.777 / 7.0$
$= 0.254$ A

CCM is maintained above 0.254 A — still well below the 2 A nominal load. The CCM verification from Deliverable 1.5 remains valid with improved margin.

Updated output capacitor requirement at 350 kHz:

$C_{out_min} = (I_{out} \times D) / (\Delta V_{out} \times f_{sw})$
$= (3 \times 0.333) / (0.05 \times 350 \times 10^3)$
$= 0.999 / 17{,}500$
$= 57.1$ μF

The minimum capacitance drops from 100 μF to 57 μF at the higher frequency. The selected 2 × 100 μF configuration from Deliverable 1.6 now provides even greater margin.

5.3 Revised Design Parameters

The following table summarises the updated design parameters incorporating the IC selection:

Parameter	Original (Deliverable 1.4)	Revised (with TPS61088)	Notes
Switching frequency	200 kHz	350 kHz	Within TPS61088 range; set by R_T resistor
Inductance	15 μH	10 μH	Within TPS61088 recommended range
ΔI_L (worst case)	0.9 A (30%)	0.76 A (25.4%)	Slightly lower — acceptable
CCM boundary	0.296 A	0.254 A	Improved
C_out minimum (3 A)	100 μF	57 μF	Relaxed; existing selection still valid
Peak I_L (3 A load)	4.95 A	4.88 A	Marginally lower

This iterative refinement — where the IC selection feeds back into the component sizing — is a normal and expected part of power supply design. The initial calculations in Deliverables 1.2–1.7 established the correct order of magnitude and design space; the IC selection now constrains the specific operating point within that space.

5.4 Why Not the LM5122?

The LM5122 is an excellent controller and would be the preferred choice for a higher-power or more flexible design. However, for this project:

- The external MOSFET requirement adds significant BOM complexity, PCB layout difficulty, and debugging surface area. For a first prototype on a student timeline, this risk is not justified when an integrated solution exists.
- The wide input/output range (up to 65 V / 100 V) is massive over-specification for a 12 V / 3 A design.
- The educational value of MOSFET selection and gate drive design is acknowledged, but the project's primary learning objectives centre on the boost converter topology, control loop behaviour, and measurement methodology — not power semiconductor selection.

If the project scope were extended to higher voltages or currents, or if the TPS61088 proved thermally limited during testing, the LM5122 would be the natural upgrade path.

5.5 Why Not the XL6009?

The XL6009E1 is **disqualified** on a hard technical constraint: its 4 A maximum switch current is exceeded by the 4.95 A peak inductor current at worst case (V_in = 8 V, I_out = 3 A). The converter would enter current limiting before reaching the specified maximum load, failing to meet the design specification.

Additionally, the non-synchronous rectification, fixed switching frequency, absence of simulation models, and poor documentation make it unsuitable for a rigorous engineering project, despite its low cost and ease of prototyping.

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6. TPS61088 Application Circuit Overview

The TPS61088 requires the following external components:

Component	Function	Preliminary Value
L (inductor)	Energy storage	10 μH, ≥ 6 A saturation, low DCR
C_out (output capacitors)	Output ripple filtering	2 × 100 μF MLCC, ≥ 16 V
C_in (input capacitor)	Input decoupling	22 μF MLCC, 25 V
R_T (timing resistor)	Sets switching frequency to 350 kHz	Per datasheet Eq. (value TBD from f_sw vs R_T curve)
R_ILIM (current limit resistor)	Sets peak switch current limit	Per datasheet Eq. (set ≥ 6 A for margin)
R_SS (soft-start resistor)	Sets soft-start time	Per datasheet Eq. (target ~5 ms)
R_FB1, R_FB2 (feedback divider)	Sets output voltage to 12 V	Per V_out = 0.6 V × (1 + R_FB1/R_FB2)
C_VCC (VCC bypass)	Internal regulator decoupling	1 μF MLCC
C_BST (bootstrap capacitor)	High-side gate drive supply	100 nF MLCC

The total external component count is approximately 12–15 components (excluding test points and optional features), which is a manageable BOM for a student PCB design.

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7. Datasheet References

Document	Identifier	Source
TPS61088 Datasheet	SLVSC30D (Rev. D, Aug 2021)	Texas Instruments

TPS61088 EVM User Guide	TPS61088EVM-638	Texas Instruments
LM5122 Datasheet	SNVS954H (Rev. H, Jun 2017)	Texas Instruments
XL6009E1 Datasheet	Rev 1.1	XLSEMI

8. Conclusion

The Texas Instruments TPS61088 is selected as the boost converter IC. Its fully integrated synchronous power stage, adjustable switching frequency, 10 A switch current capability, peak current mode control, and excellent documentation ecosystem make it the optimal choice for a 12 V / 3 A boost converter within a final year project timeline.

The IC selection necessitates an iterative update to the switching frequency (from 200 kHz to 350 kHz) and inductance (from 15 μH to 10 μH) to comply with the TPS61088's recommended inductor range. All prior deliverable conclusions — CCM verification, output capacitor sizing, and input capacitor sizing — remain valid with improved margin at the revised operating point.

The next deliverables (1.9–1.11) will address the specific component selection for the inductor, capacitors, and feedback network using the TPS61088's datasheet equations and recommended component families.